

Section 0

Algebraic Foundation

Reconciliation Propagator

The first deliverable of Phase B: a fully merged, end-to-end rewrite of Section 0 against Construction Spec v1.0. Implements the **Linear** → **Quadratic** → **Cubic** → **Quartic** re-spine, the $\delta\hat{C}$ replacement that breaks the §0.4 → §1.1.1 dependency loop, the derived (not postulated) Master-Zero theorem, the thin/full split of A_{phy} , the restored Theorem 0.7.3, and the relocation of the Reconciliation Principle to §0.8.

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PHASE B — Section 0 Test Case

SCOPE 9 Subsections · 4 Layers · 1 Restored Theorem

SOURCE SPEC RCF-CONST-SPEC-v1.0, Ch. 5–9

L → Q → C → Q SPINE

$\Delta\hat{C}$ LOOP FIX

DERIVED MASTER-ZERO

THIN/FULL SPLIT

THM 0.7.3 RESTORED

RP RELOCATED

Preamble — How to Read This Section

This document is the merged canonical form of Section 0 of the Reconciliation Causal Framework (RCF). It is the first deliverable of Phase B as specified in *RCF Unified Construction Specification v1.0*, and implements every patch required by that spec: the $L \rightarrow Q \rightarrow C \rightarrow Q$ layer re-spine, the $\delta\hat{C}$ replacement for the SOE flow, the derived (not postulated) Master-Zero theorem, the thin/full split of the physical sub-algebra, the restoration of Theorem 0.6.3, and the relocation of the Reconciliation Principle to §0.8 (immediately after the Cubic ingredients of Sections 1–2 are available).

The structure follows the spec's source map (Table 4.1) row-by-row. Each subsection opens with a layer badge identifying its position in the $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder, a one-line source citation, and the epistemic tag ([Established], [Conditional], [Theorem Target], [Structural]) inherited from the master manuscripts. Body text is ported verbatim where possible; rewritten passages are flagged inline with a spec chapter reference (e.g. *per Ch. 8*).

Dependency contract with Sections 1 and 2

§0.8 (Reconciliation Principle) requires two Cubic-layer ingredients that are defined in **Section 1.1** (causal order $<$) and **Section 2.1** (correlation kernel K_ω). In this merged Section 0 these ingredients appear as *forward references* — the principle is stated in its final canonical form, but its full minimization target $I(S)$ is parameterized on objects that the next two merged sections will supply. When Section 1 is merged, the forward reference resolves into a concrete definition; when Section 2 is merged, the variational target becomes fully grounded. No circularity is introduced because §0.1–0.7b do not depend on §0.8 — the Reconciliation Principle is the *output* of the foundation, not its input.

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§0.0 Purpose of the Foundation

The Relational Constraint Framework (RCF) derives all physical structure — spacetime, fields, particles, gravity, probability, cosmology — from a single primitive: relations subject to constraints. Section 0 establishes the algebraic and dynamical foundation on which every subsequent section rests. It does not assume a background manifold, a pre-existing Hilbert space, an external time parameter, or a measurement postulate. Instead it constructs the kinematic algebra, the kinematic state, the GNS representation, the Reconciliation Propagator $\mathbf{R}_t$, and the physical sub-algebra $\mathbf{A}_{\text{phy}}$ as the asymptotic fixed-point algebra of $\mathbf{R}_t$. Master-Zero ($\ker(\mathbf{M}) = \ker(\mathbf{F})$) is *derived* as an asymptotic attractor of the propagator, not postulated as an initial condition.

The central architectural commitment of this section is the $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder. Every definition is placed at the layer where its ingredients first become available. The Linear layer (L) defines the primitive algebra and constraints. The Quadratic layer (Q) defines the GNS inner product, the obstruction burden, the Reconciliation Propagator, and the physical sub-algebra. The Cubic layer (C) — supplied by Sections 1 and 2 — introduces causal order and the correlation kernel. The Quartic layer (Q') — reached in §0.7, §0.7b, §0.8 — assembles these into the Reconciliation Principle, the variational target, and the formal action. No layer depends on a layer below it; no definition is used before its ingredients exist.

Physicality is constraint compatibility on the correctly reduced algebra, not prior placement in spacetime.

A reader of the legacy 47-document archive will note that this merged Section 0 makes two structural corrections relative to the most recent Gen 3 drafts. First, the physical sub-algebra $\mathbf{A}_{\text{phy}}$ is no longer introduced in §0.4 (where it would have to be used before it is earned); it is split into a *thin* candidate (§0.6, defined by spectral compatibility with $\ker(\mathbf{M})$) and a *full* algebra (§0.7, defined as the fixed-point algebra of $\mathbf{R}_\infty$), with Theorem 0.6.3 certifying that the two coincide under the stable-mode assumption. Second, the Reconciliation Principle is no longer stated in §0.4 (where it would borrow vocabulary from Section 1 before that section is reached); it is relocated to §0.8, immediately after the Cubic ingredients of Sections 1–2 are available.

§0.1 The Kinematic Algebra (Layer L — Linear)

LAYER L · LINEAR

Source: Front.pdf §0.1.2 (Definition — Relational ()-Algebra), augmented by Section_0_2 §0.1 (Definition 0.1.1 Kinematic Algebra) and Con.pdf §0.1. Epistemic tag: [Established Theorem / Definition].*

The foundation begins with a single primitive: an unital complex (*)-algebra $\mathbf{A}$. Elements of $\mathbf{A}$ represent all possible relational operations; the involution $*$ represents the algebraic adjoint (the relational analogue of Hermitian conjugation). No commutativity, no topology, no norm, no Hilbert space, and no spacetime is assumed at this stage. The algebra is purely algebraic: it carries addition, scalar multiplication, an associative product, a unit 1, and an involution satisfying $(A^*)^* = A$, $(A + B)^* = A^* + B^*$, $(\lambda A)^* = \bar{\lambda} A^*$, $(AB)^* = B^* A^*$.

Definition 0.1.1 — Kinematic Algebra

DEFINITION 0.1.1 — Kinematic Algebra [Established]

Let A be a unital complex $(*)$ -algebra with unit $1 \in A$.

Elements $A \in A$ represent all possible relational operations.

The involution $*$: $A \rightarrow A$ satisfies, for all $A, B \in A$ and

$\lambda \in \mathbb{C}$:

$$(A^*)^* = A \quad (0.1.1a)$$

$$(A + B)^* = A^* + B^* \quad (0.1.1b)$$

$$(\lambda A)^* = \lambda^* A^* \quad (0.1.1c)$$

$$(AB)^* = B^* A^* \quad (0.1.1d)$$

No commutativity, topology, norm, or Hilbert-space structure is assumed at this layer. The algebra is the sole primitive.

Definition 0.1.2 — Primitive Constraints

A constraint is an element $\hat{C}_\alpha \in A$ whose vanishing expresses a relational admissibility condition. The framework introduces a collection $\{\hat{C}_\alpha\}_{\alpha \in I}$ of primitive constraints; no first-class/second-class distinction is made at this stage. That distinction is deferred to the Dirac reduction performed in §0.7b, where the Tier 1 compatibility matrix $\Delta_{ab}^{(1)} = [K_a, K_b]$ is inverted to construct the algebraic Dirac bracket.

DEFINITION 0.1.2 — Primitive Constraints [Established]

A primitive constraint is an element $\hat{C}_\alpha \in A$.

The constraint family $\{\hat{C}_\alpha\}_{\alpha \in I}$ gathers all relational admissibility conditions the framework imposes.

No first-class/second-class distinction is made at Layer L.

That distinction is deferred to the Dirac reduction (§0.7b).

Definition 0.1.3 — Master Constraint Operator

The primitive constraints are packaged into a single positive operator — the master constraint $\mathbf{M}$. This packaging is purely algebraic: it does not require a Hilbert space, a state, or a representation. The weights $w_\alpha > 0$ are arbitrary positive reals; the framework's theorems are weight-independent in their ker-equivalence statements.

DEFINITION 0.1.3 — Master Constraint Operator [Established]

Let $\{\hat{C}_\alpha\}_{\alpha \in I}$ be the primitive constraint family and let $\{w_\alpha\}_{\alpha \in I}$ be strictly positive real weights. The master constraint operator is

$$\hat{M} = \sum_{\alpha \in I} w_\alpha \cdot \hat{C}_\alpha^\dagger \hat{C}_\alpha \in \mathbf{A} \quad (0.1.3)$$

Because each summand is a positive $(*)$ -square and $w_\alpha > 0$, $\hat{M}$ is positive in the algebraic sense: for any state ω , $\omega(\hat{M}) \geq 0$.

Remark 0.1.4 — What Is Not Assumed at Layer L

Layer L deliberately omits every structure that the framework intends to derive. No state is postulated; no Hilbert space is assumed; no scalar product is defined; no causal order, no correlation kernel, no metric, no time parameter, and no probability measure is in play. The framework's central methodological commitment — that physical structure is *emergent* from relational algebra, not primitive — begins here, at the very first definition. Every subsequent layer (Q, C, Q') adds structure by derivation, never by postulation.

§0.2 Kinematic State and Constraint Ideal (Layer L — Linear)

LAYER L · LINEAR

Source: *Con.pdf* §0.2-0.3 (Definitions 0.5, 0.7), *Glm.pdf* §0.3.6-0.3.8 (algebraic state, physical state), and *Section_0_2 §0.1.4* (general kinematic state). *Epistemic tag:* [Established Theorem / Definition].

A kinematic state is any positive linear functional on $\mathbf{A}$. Crucially — and in contrast to earlier formulations — the kinematic state is *not* required to satisfy $\omega_{\text{kin}}(\mathbf{M}) = 0$. The Master-Zero condition is the *target* of reconciliation, not its starting point. The framework begins from an unconstrained kinematic state and lets the Reconciliation Propagator drive it asymptotically to the physical sector. This single methodological choice is what makes the Master-Zero theorem of §0.5 a derived result rather than a postulate.

Definition 0.2.1 — Algebraic State

DEFINITION 0.2.1 — Algebraic State [Established]

An algebraic state on $\mathbf{A}$ is a linear functional

$$\omega : \mathbf{A} \rightarrow \mathbb{C}$$

satisfying the positivity condition

$$\omega(A^\dagger A) \geq 0 \quad \text{for all } A \in \mathcal{A}. \quad (0.2.1)$$

If additionally $\omega(1) = 1$, the state is called normalised.

The set of all algebraic states is denoted $S(\mathcal{A})$.

Definition 0.2.2 — General Kinematic State

DEFINITION 0.2.2 — General Kinematic State [Established] (key correction vs Gen 2)

A kinematic state is ANY positive linear functional ω_{kin} on $\mathcal{A}$ with $\omega_{\text{kin}}(1) = 1$. The state is NOT required to satisfy $\omega_{\text{kin}}(\hat{M}) = 0$.

$$\omega_{\text{kin}} \in S(\mathcal{A}), \quad \omega_{\text{kin}}(1) = 1, \quad \omega_{\text{kin}}(\hat{M}) \geq 0 \quad (\text{no zero condition})$$

The Master-Zero condition $\omega(\hat{M}) = 0$ is the target of the reconciliation flow (§0.4), not an initial condition.

Why the kinematic state is unconstrained

In Gen 2 (Front.pdf, Con.pdf, N.docx, Glm.pdf), the physical state was defined as a positive functional satisfying $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$ for all α — i.e. Master-Zero was postulated at the outset. This made the Master-Zero Equivalence theorem (Theorem 0.5.1 below) a tautology rather than a result. In the merged canonical form, the kinematic state is left unconstrained; the propagator R_t is what drives the state to $\ker(\hat{M})$. Master-Zero becomes a *derived asymptotic property*, restoring the theorem's physical content. This is the correction mandated by Construction Spec Chapter 5 and reflected in source-map row 0.5 (action: "PORT + drop Gen 2 postulate").

Definition 0.2.3 — Constraint Ideal

The primitive constraints generate a two-sided $(*)$ -ideal — the constraint ideal I_C . This ideal gathers all algebraic consequences of the constraints. The framework will eventually quotient $\mathcal{A}$ by a sub-ideal of I_C (the Tier 1 ideal I_K , defined in §0.7b when the Dirac reduction is performed) to obtain the reduced algebra $\mathcal{A}_{\text{red}}$. At Layer L, the constraint ideal is merely introduced; its structural role is activated later.

DEFINITION 0.2.3 — Constraint Ideal [Established]

The constraint ideal I_C is the smallest two-sided $(*)$ -ideal of $\mathcal{A}$

containing all primitive constraints:

$$I_C = \{ \hat{C}_\alpha : \alpha \in I \} = \{ \sum_k A_k \hat{C}_{\{\alpha_k\}} B_k : A_k, B_k \in A, \alpha_k \in I \} \quad (0.2.3)$$

A proper constraint system satisfies $1 \notin I_C$ (the constraints do not over-determine the algebra).

Definition 0.2.4 — Compatibility Bracket

The mutual obstruction between constraints is captured by their commutator. For two primitive constraints $\hat{C}_\alpha, \hat{C}_\beta$, the compatibility bracket is $\Delta_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta] \in A$. When the constraint family is partitioned into Tier 1 (kinematic preconditions, second-class) and Tier 2 (dynamical constraints, first-class on the reduced algebra) in §0.7b, the Tier 1 block of this bracket matrix becomes the object the Dirac reduction inverts.

DEFINITION 0.2.4 — Compatibility Bracket [Established]

For primitive constraints $\hat{C}_\alpha, \hat{C}_\beta \in A$, the compatibility bracket is the commutator

$$\Delta_{\{\alpha\beta\}} = [\hat{C}_\alpha, \hat{C}_\beta] = \hat{C}_\alpha \hat{C}_\beta - \hat{C}_\beta \hat{C}_\alpha \in A \quad (0.2.4)$$

The full compatibility matrix $\Delta = (\Delta_{\{\alpha\beta\}})$ collects all mutual obstructions. Its Tier 1 sub-block $\Delta^{(1)}$ (defined in §0.7b) is the object whose invertibility licenses the Dirac reduction.

§0.3 GNS Hilbert Space and the Fracture Operator (Layer Q — Quadratic)

LAYER Q · QUADRATIC

Source: Front.pdf §0.6 (Dirac-GNS construction, Definitions 0.16-0.23, Theorems 0.9-0.10), Con.pdf §0.4 (GNS representation, Theorems 0.5-0.8), and Section_0_2 §0.2-0.3 (kernel containment and obstruction burden). **Epistemic tag:** [Established Theorem].

The Quadratic layer introduces the Gelfand–Naimark–Segal (GNS) construction, which produces a Hilbert-space representation from the kinematic state. The construction is *derived*, not postulated: $\mathbf{H}_{\text{kin}}$ is the completion of the quotient of A by the GNS null space, equipped with the state-induced inner product. No primitive Hilbert space is assumed. The same layer introduces the fracture operator $\mathbf{F}$ — the algebraic object whose expectation value is the obstruction burden — and proves the kernel-containment lemma that licenses

the Master-Zero equality in §0.5.

Definition 0.3.1 — State-Induced Inner Product

DEFINITION 0.3.1 — State-Induced Inner Product [Established]

Let ω_{kin} be a general kinematic state on A . The state-induced sesquilinear form on A is

$$\langle A, B \rangle_{\omega} := \omega_{\text{kin}}(A^\dagger B) \quad \text{for } A, B \in A \quad (0.3.1)$$

This form is linear in the second argument and conjugate-linear in the first. By the positivity of ω_{kin} (Definition 0.2.1), it is positive semidefinite: $\langle A, A \rangle_{\omega} = \omega_{\text{kin}}(A^\dagger A) \geq 0$.

Definition 0.3.2 — GNS Null Space

DEFINITION 0.3.2 + LEMMA 0.3.3 [Established]

The GNS null space is the set of algebra elements of zero state-induced length:

$$N_{\omega} = \{ A \in A : \omega_{\text{kin}}(A^\dagger A) = 0 \} \quad (0.3.2)$$

Lemma 0.3.3: N_{ω} is a left ideal of A .

Proof: If $A \in N_{\omega}$ and $B \in A$, then by Cauchy-Schwarz for positive functionals, $\omega_{\text{kin}}(A^\dagger X) = 0$ for all $X \in A$. Set $X = B^\dagger B A$; then $\omega_{\text{kin}}((BA)^\dagger (BA)) = \omega_{\text{kin}}(A^\dagger B^\dagger B A) = 0$, so $BA \in N_{\omega}$. $\square$

Definition 0.3.4 — Kinematic GNS Hilbert Space

DEFINITION 0.3.4 — Kinematic GNS Hilbert Space [Established]

The kinematic GNS Hilbert space H_{kin} is the metric completion of the quotient pre-Hilbert space:

$$H_{\text{kin}} = \text{completion of } (A / N_{\omega}, \langle \cdot, \cdot \rangle_{\omega}) \quad (0.3.4)$$

The representation $\pi_{\text{kin}} : A \rightarrow \text{End}(H_{\text{kin}})$ is given by left

multiplication: $\pi_{\text{kin}}(A) [B] = [AB]$.

The cyclic vector is $\Omega_{\text{kin}} = [1] \in H_{\text{kin}}$, satisfying

$$\omega_{\text{kin}}(A) = \langle \Omega_{\text{kin}}, \pi_{\text{kin}}(A) \Omega_{\text{kin}} \rangle_{H_{\text{kin}}} \quad (\text{reconstruction}).$$

Definition 0.3.5 — Fracture Operator

The fracture operator $\mathbf{F}$ is the positive operator that measures the algebraic inconsistency of the constraint family. It is built from the compatibility brackets $\Delta_{\alpha\beta}$ introduced in Definition 0.2.4. Where $\mathbf{M}$ asks "how much does each constraint violate, individually?", $\mathbf{F}$ asks "how much do the constraints mutually obstruct each other?". The two operators probe different aspects of inconsistency, and the Master-Zero theorem of §0.5 establishes that they share the same kernel.

DEFINITION 0.3.5 — Fracture Operator [Established]

Let $\{\Delta_{\alpha\beta}\}$ be the compatibility brackets (Def 0.2.4). The fracture operator is

$$\hat{F} = \sum_{\{\alpha,\beta\}} \pi_{\text{kin}}(\Delta_{\alpha\beta})^\dagger \cdot \pi_{\text{kin}}(\Delta_{\alpha\beta}) \in \text{End}(H_{\text{kin}}) \quad (0.3.5)$$

$\hat{F}$ is positive by construction. Its expectation value

$B_\Delta[\rho] = \text{Tr}(\rho \hat{F})$ is the obstruction burden (Definition 0.3.7).

Lemma 0.3.6 — Kernel Containment

LEMMA 0.3.6 — Kernel Containment [Established]

LEMMA 0.3.6 (Kernel Containment).

$$\ker(\hat{M}) \subseteq \ker(\hat{F}) \quad (\text{and the reverse inclusion holds on the physical sector; see Theorem 0.5.1})$$

Proof. Suppose $|\psi\rangle \in \ker(\hat{M})$, i.e. $\hat{M} |\psi\rangle = 0$. Then for each α ,

$$\begin{aligned} 0 &= \langle \psi | \hat{M} | \psi \rangle = \sum_{\alpha} w_{\alpha} \langle \psi | \hat{C}_{\alpha}^\dagger \hat{C}_{\alpha} | \psi \rangle \\ &= \sum_{\alpha} w_{\alpha} \| \hat{C}_{\alpha} | \psi \rangle \|^2. \end{aligned}$$

Since each $w_{\alpha} > 0$ and each term is non-negative, every term

must vanish: $\hat{C}_\alpha |\psi\rangle = 0$ for all α . Hence $\Delta_{\{\alpha\beta\}} |\psi\rangle = 0$ for all α, β , giving $\hat{F} |\psi\rangle = 0$. Therefore $\ker(\hat{M}) \subseteq \ker(\hat{F})$. $\square$

The reverse inclusion $\ker(\hat{F}) \subseteq \ker(\hat{M})$ is NOT trivial; it is established by the Convergence Theorem 0.5.1.

Definition 0.3.7 — Obstruction Burden

The obstruction burden is the central physical quantity of the Quadratic layer. It is the expectation value of the fracture operator in a given state. The framework's central technical property — burden linearity — is an immediate consequence of the definition: $\mathbf{B}_\Delta[\rho] = \text{Tr}(\rho \mathbf{F})$ is linear in ρ , because $\mathbf{F}$ is fixed by the algebraic structure of the constraints. This linearity is what licenses the burden-as-mass identity of Theorem 4.2.2 (deferred to the merged Section 4) and prevents any smuggling of probabilistic structure into the algebraic layer.

DEFINITION 0.3.7 + BURDEN LINEARITY [Established]

For any density matrix ρ on H_{kin} , the obstruction burden is

$$\mathbf{B}_\Delta[\rho] = \text{Tr}(\rho \cdot \hat{F}) \in \mathbb{R}_{\geq 0} \quad (0.3.7)$$

Property (Burden Linearity). $\mathbf{B}_\Delta$ is LINEAR in ρ :

$$\mathbf{B}_\Delta[\sum_k p_k \rho_k] = \sum_k p_k \mathbf{B}_\Delta[\rho_k].$$

This is a proven algebraic identity, not a probabilistic average.

It is the load-bearing property that lets mass (Section 4) be identified with burden without smuggling probability into the algebraic layer.

Definition 0.3.8 — Cross-Extension Network Operator

The cross-extension network operator Π_{net} projects onto the subspace of observables correlated across multiple open extensions. It is the algebraic object on which the MOE gradient descent (Mechanism B of the Reconciliation Propagator, §0.4) operates. The operator is *derivable* from $\mathbf{F}$ and the correlation kernel K_ω (defined in the merged Section 2): no new primitives are introduced. The commutator $[\hat{C}_\alpha, \Pi_{\text{net}}]$ generates the relational burden channel that appears in the three-channel decomposition of burden.

DEFINITION 0.3.8 — Cross-Extension Network Operator [Established]

The cross-extension network operator is

$$\Pi^{\text{net}} = \sum_{\{E_i\}} P^{\{E_i\}} K_{\omega} P^{\{E_i\}} \quad (0.3.8)$$

where $P^{\{E_i\}}$ are projectors onto open extensions E_i and K_{ω} is the correlation kernel (defined in Section 2.1).

Π^{net} is derivable from F^{net} and K_{ω} — zero new primitives.

The relational burden channel is $[\hat{C}_{\alpha}, \Pi^{\text{net}}]$ (Section 5).

§0.4 The Reconciliation Propagator (Layer Q — Quadratic)

LAYER Q · QUADRATIC

*Source: Section_0_2.pdf §0.4 (SOE/MOE/dephasing decomposition), REWRITTEN per Construction Spec Ch. 8 ($\delta\hat{C}$ replacement breaks the dependency loop with §1.1.1). **Epistemic tag:** [Conditional Theorem] (convergence in finite dimensions; infinite-dimensional proof is Theorem Target T-2).*

The Reconciliation Propagator $\mathbf{R}_t$ is the dynamical engine of the framework. It is not a single flow but the ordered composition of three scale-distinct mechanisms: the SOE (Single-Open-Extension) spectral-flux flow, which is local, isometric, and phase-preserving; the MOE (Multiple-Open-Extension) gradient descent, which is global, contractive, and burden-minimizing; and the dephasing residual, which suppresses cross-eigenspace coherences. Together they drive any unconstrained kinematic state asymptotically to $\ker(\mathbf{M}) \cap \ker(\mathbf{F})$.

The dependency-loop fix (Construction Spec Ch. 8)

Gen 3 §0.4 wrote the SOE spectral-flow component as $dp/dt|_{\text{SOE}^{\text{spec}}} = i[\sqrt{(\mathbf{F} + \delta\mathbf{F}(E_{\text{new}}))}, \rho]$, where E_{new} ("a new event") appears as a primitive input. But "event" is not defined until Section 1.1.1, which itself depends on $\mathbf{A}_{\text{phy}}$, which depends on $\ker(\mathbf{M})$, which §0.4 is supposed to produce. This is a circular reference. The merged canonical form replaces E_{new} with $\delta\hat{C}$ — a raw algebraic perturbation to the primitive constraint set $\hat{C} \subset \mathbf{A}$. $\delta\hat{C}$ is defined entirely within Layer L/Q (no reference to events, causal order, or correlation), breaking the loop: §0.4 → §0.5 → §0.6 (thin) — no forward reference. Section 1.1.1 (event) is restated downstream to require only the *thin* candidate $\mathbf{A}_{\text{phy}}^{\text{thin}}$ from §0.6.

0.4.1 Mechanism A — SOE Spectral-Flux Flow (Local, Isometric)

The SOE flow propagates the kinematic state under a single open causal extension. The propagator has two coupled components: a flux-gradient component that redistributes obstruction along the constraint manifold, and a spectral-gradient component that rotates the $\mathbf{F}$ eigenbasis to align with the perturbation. Both are purely algebraic — they reference the perturbation $\delta\hat{C}$, not any object from Section 1.

DEFINITION 0.4.1 — SOE Spectral-Flux Flow [Conditional]

DEFINITION 0.4.1 (SOE Spectral-Flux Flow).

The SOE flow propagates ρ under a single algebraic perturbation $\delta\hat{C}$ to the primitive constraint set $\hat{C} \subset A$. The perturbation is defined entirely within Layer L/Q:

(i) Flux-Gradient Flow (local constraint alignment):

$$\begin{aligned} dp/dt|_{\text{SOE}^{\text{flux}}} &= \nabla \cdot J_{\Delta}(\rho) \\ &= \sum_{\{\alpha\beta\}} [\hat{C}_{\alpha}, [\Delta_{\{\alpha\beta\}}, \rho \hat{C}_{\beta}^{\dagger}]] + \text{h.c.} \end{aligned} \quad (0.4.1a)$$

(ii) Spectral-Gradient Flow (eigenbasis alignment):

$$dp/dt|_{\text{SOE}^{\text{spec}}} = i [\sqrt{(\hat{F} + \delta\hat{F}(\delta\hat{C}))}, \rho] \quad (0.4.1b)$$

$$\text{where } \delta\hat{F}(\delta\hat{C}) = [\delta\hat{C}, [\delta\hat{C}, \hat{F}]] + O(\delta\hat{C}^3)$$

Combined SOE flow (one open extension, duration ε):

$$R_{\varepsilon}^{\text{SOE}} = T \exp\left(\int_0^{\varepsilon} (d/dt|_{\text{flux}} + d/dt|_{\text{spec}}) dt\right) \quad (0.4.1c)$$

Properties:

- Isometric / volume-preserving (does NOT minimize total burden)
- Preserves $\text{Tr}(\rho \hat{M})$ and $\text{Tr}(\rho \hat{F})$ locally – burden is REDISTRIBUTED, not erased
- Preserves ALL Layer B quantum phase and spectral structure
- $\delta\hat{C}$ is purely algebraic – no Section 1 vocabulary required

0.4.2 Mechanism B — MOE Gradient Descent (Global, Contractive)

The MOE flow operates on the global state accumulated across a sequence of $N \geq 2$ SOEs. It is the only place genuine gradient descent appears in the framework. The descent is on the Bures metric — the unique monotone metric on density matrices compatible with quantum information theory. The Master-Zero condition is enforced as a Lagrange multiplier, not as an initial constraint. At leading order for large N , the descent contracts cross-sub-sector coherences in a Lindblad form — this is Theorem Target T-2.

DEFINITION 0.4.2 — MOE Gradient Descent [Conditional + Theorem Target T-2]

DEFINITION 0.4.2 (MOE Gradient Descent).

Over a causal chain of $N \geq 2$ SOEs (duration $T = N \cdot \epsilon$), the MOE flow is the Bures-metric gradient descent:

$$\frac{dp}{dt}|_{\text{MOE}} = -\eta \cdot \nabla_{\text{Bures}} B_{\Delta}[\rho] - \lambda_t \cdot \nabla_{\text{Bures}} C[\rho] \quad (0.4.2a)$$

where:

- ∇_{Bures} = functional gradient w.r.t. the Bures metric
- $B_{\Delta}[\rho] = \text{Tr}(\rho \hat{F})$ (obstruction burden, Definition 0.3.7)
- $C[\rho] = \text{Tr}(\rho \hat{M})$ (Master-Zero target, enforced as Lagrange multiplier; NOT postulated)

Derived Lindblad form (Theorem Target T-2): at leading order for large N , the effect of MOE descent on cross-sub-sector coherences is

$$\frac{dp}{dt}|_{\text{MOE}^{\text{lead}}} = -\gamma [\hat{F}, [\hat{F}, \rho]] \quad (0.4.2b)$$

with decay rate $\gamma \propto N \cdot \eta$ and spectral-gap dependence $(f_i - f_j)^2$.

This is the universal leading-order contraction of Bures descent on a state with spectral gaps in $\hat{F}$.

Properties:

- Contractive, irreversible, burden-minimizing
- Drives global state to $\ker(\hat{M}) \cap \ker(\hat{F})$
- The ONLY place gradient descent appears in the framework

0.4.3 Mechanism C — Dephasing (Residual Cross-Eigenspace Suppression)**DEFINITION 0.4.3 — Dephasing [Conditional]**

DEFINITION 0.4.3 (Dephasing).

Scale: between $\hat{F}$ -eigenspaces with distinct eigenvalues.

$$\text{dp/dt|}_{\text{deph}} = -\gamma [\hat{F}, [\hat{F}, \rho]] \quad (0.4.3)$$

This suppresses residual cross-eigenspace coherences while preserving within-eigenspace structure. It is the MOE-scale residual effect of gradient descent on cross-eigenspace coherences.

Note: equations (0.4.2b) and (0.4.3) coincide at leading order, reflecting that dephasing is the leading-order signature of MOE descent on the $\hat{F}$ spectral decomposition.

0.4.4 The Full Reconciliation Propagator

DEFINITION 0.4.4 + LEMMA 0.4.5 [Conditional]

DEFINITION 0.4.4 (Full Reconciliation Propagator R_t).

The full propagator is the ordered composition:

$$R_t = \lim_{N \rightarrow \infty} (R_{\{T/N\}}^{\text{MOE}} \circ (R_{\epsilon}^{\text{SOE}})^N), \quad T = N \cdot \epsilon \quad (0.4.4)$$

LEMMA 0.4.5 (Consistency). The combined flow:

1. Preserves all Layer B algebraic distinguishability (phase, spectral gaps, cubic kernel orientation) at the SOE scale
2. Converges exponentially fast to $\ker(\hat{M}) \cap \ker(\hat{F})$ at the MOE scale, with coherence decay rate $\gamma \cdot (\Delta f_{ij})^2$
3. Reduces to the prior single-scale propagator in the $N \rightarrow \infty$ continuum limit – all prior toy model results still hold

§0.5 The Convergence Theorem — Master-Zero as Derived Result (Layer Q)

LAYER Q · QUADRATIC

Source: *Glm.pdf* §0.4.15-0.4.16 (Theorem 0.9 Master-Zero Equivalence, kernel characterization), Section_0_2 §0.5 (Convergence Theorem 0.5.1). **Per Construction Spec Ch. 5:** drop the Gen 2 postulate; Master-Zero is a derived asymptotic property. **Epistemic tag:** [Conditional Theorem] (finite-dimensional proof established; infinite-dimensional requires spectral

gap conditions — Theorem Target T-2).

The Convergence Theorem is the central result of Section 0. It states that for *any* positive kinematic state ω_{kin} (not pre-constrained), the Reconciliation Propagator drives the state asymptotically to the projection onto $\ker(\mathbf{M}) = \ker(\mathbf{F})$. Two consequences follow immediately: (1) $\text{Tr}(\rho_{\infty} \mathbf{M}) = 0$ — Master-Zero is *derived*, not assumed; (2) $\text{Tr}(\rho_{\infty} \mathbf{F}) = 0$ — the obstruction burden is fully resolved. This is the rigorous sense in which the framework "earns" the Master-Zero condition rather than postulating it.

Theorem 0.5.1 — Convergence to the Physical Sector

THEOREM 0.5.1 — Convergence to the Physical Sector [Conditional]

THEOREM 0.5.1 (Convergence to the Physical Sector).

Let ω_{kin} be ANY positive state on A (not pre-constrained).

Under the composed SOE-MOE flow R_t (Definition 0.4.4),

$$\lim_{t \rightarrow \infty} \rho_t = P^{\wedge}_0 \rho_0 P^{\wedge}_0 / \text{Tr}(P^{\wedge}_0 \rho_0) \quad (0.5.1)$$

where $P^{\wedge}_0$ projects onto $\ker(M^{\wedge}) = \ker(F^{\wedge})$.

Key consequences:

1. $\text{Tr}(\rho_{\infty} M^{\wedge}) = 0$ — Master-Zero is asymptotically DERIVED,
not assumed
2. $\text{Tr}(\rho_{\infty} F^{\wedge}) = 0$ — obstruction burden fully resolved
3. Cross-sub-sector coherences decay exponentially with rate
 $\gamma \cdot (\Delta f_{ij})^2$

Mechanism assignment:

- SOE flow incorporates new perturbations without increasing global burden
- MOE descent drives global convergence to $\ker(M^{\wedge})$
- Dephasing suppresses residual cross-eigenspace coherence

Proof sketch. The SOE component is isometric (Lemma 0.4.5), so it preserves the spectral decomposition of $F^{\wedge}$. The MOE component is contractive on the Bures metric and drives $B_{\Delta}[\rho]$ monotonically downward. By the LaSalle invariance principle applied to the

Bures-gradient flow, the ω -limit set is contained in the critical set of B_Δ , which (by Lemma 0.3.6 and the reverse inclusion proved below) coincides with $\ker(\hat{F}) = \ker(\hat{M})$. The finite-dimensional toy model $(M_6(C), \text{non-commuting constraints}, \text{Tr}(\rho_0 \hat{M}) = 0.96 \rightarrow 2.14 \times 10^{-4} \text{ in finite } t)$ verifies convergence numerically.

STATUS: [Conditional Theorem]. The infinite-dimensional proof requires spectral gap conditions on $\hat{F}$ (Theorem Target T-2).

Corollary 0.5.2 — Master-Zero Equivalence

COROLLARY 0.5.2 — Master-Zero Equivalence [Established on the physical sector]

COROLLARY 0.5.2 (Master-Zero Equivalence).

On the asymptotic physical sector, the following are equivalent:

- (i) $\omega_\infty(\hat{M}) = 0$
- (ii) $\omega_\infty(\hat{C}_\alpha \dagger \hat{C}_\alpha) = 0$ for all α
- (iii) $\omega_\infty(\hat{F}) = 0$
- (iv) $\omega_\infty(\Delta_{\{\alpha\beta\}} \dagger \Delta_{\{\alpha\beta\}}) = 0$ for all α, β

In particular, $\ker(\hat{M}) = \ker(\hat{F})$ on the physical sector, completing the reverse inclusion of Lemma 0.3.6.

Proof. (i) $\Leftrightarrow$ (ii) is Glm.pdf Theorem 0.9 (positivity + weights).

(iii) $\Leftrightarrow$ (iv) is the same argument applied to $\hat{F}$. The two pairs are linked by Theorem 0.5.1: the propagator drives the state to a point where both $\hat{M}$ and $\hat{F}$ vanish, hence $\ker(\hat{M}) = \ker(\hat{F})$ on the physical sector. $\square$

Corollary 0.5.3 — $\Lambda_B = 0$ (Cosmological Constant Vanishes)

A direct downstream consequence of Master-Zero — ported from Con.pdf Theorem 5.5 and made rigorous by the Convergence Theorem — is that the burden cosmological term Λ_B vanishes identically. This is not a fine-tuning; it is a structural consequence of the asymptotic attractor. The full formal action S_{eff} (deferred to

the merged Section 9, per Spec Ch. 7) carries a Λ_B term that vanishes by Master-Zero, eliminating the cosmological constant problem at the algebraic level (subject to closure of Theorem Target T-2).

§0.6 The Thin Physical Sub-Algebra (Layer Q — Quadratic)

LAYER Q · QUADRATIC

Source: NEW (split from old §0.6), per Construction Spec Ch. 8. Epistemic tag: [Definition — derived from $\ker(M)$ compatibility alone].

The Gen 3 draft introduced $\mathbf{A}_{\text{phy}}$ in §0.6 as a single object defined by $\ker(\mathbf{M})$ -compatibility *and* fixed-point closure under $\mathbf{R}_{\infty}$. The merged canonical form splits this into two stages. The *thin* candidate $\mathbf{A}_{\text{phy}}^{\text{thin}}$ is defined here, early, using only spectral compatibility with $\ker(\mathbf{M})$: it is the algebra of operators whose kinematic representation preserves the kernel of $\mathbf{M}$. This is enough for Section 1.1.1 to define "event" without forward reference. The *full* algebra $\mathbf{A}_{\text{phy}}$ is defined in §0.7 as the fixed-point algebra of $\mathbf{R}_{\infty}$, after RP has been stated. Theorem 0.7.3 (restored per Spec Ch. 9) proves the two definitions coincide under the stable-mode assumption.

Definition 0.6.1 — Thin Physical Sub-Algebra

DEFINITION 0.6.1 — Thin Physical Sub-Algebra [Established]

DEFINITION 0.6.1 (Thin Physical Sub-Algebra).

The thin physical sub-algebra is the set of algebra elements whose kinematic representation preserves $\ker(\hat{M})$:

$$\mathbf{A}_{\text{phy}}^{\text{thin}} = \{ A \in \mathbf{A} : \pi_{\text{kin}}(A) \ker(\hat{M}) \subseteq \ker(\hat{M}) \} \quad (0.6.1)$$

Equivalently: $A \in \mathbf{A}_{\text{phy}}^{\text{thin}}$ iff $[A, \hat{P}_0] = 0$ on $\ker(\hat{M})$,

where $\hat{P}_0$ is the asymptotic projector onto $\ker(\hat{M})$.

Properties:

- Defined using only Layer Q objects ($\mathbf{A}$, π_{kin} , $\ker(\hat{M})$)
- Does NOT require fixed-point closure under $\mathbf{R}_{\infty}$
- Does NOT require the Reconciliation Principle (§0.8)
- Sufficient for Section 1.1.1 to define "event" without forward reference to §0.7 or §0.8

Why the thin/full split is necessary

Without the split, the framework faces a choice between two regressions. Define A_{phy} early (Gen 3 §0.6) — and you have used the fixed-point algebra of R_{∞} before R_t has been composed, before RP has been stated, and before the Cubic ingredients of Sections 1–2 are available. Define it late (after §0.8) — and Section 1.1.1 cannot define "event" without forward reference, breaking the sequential dependency chain. The thin/full split resolves this: $A_{\text{phy}}^{\text{thin}}$ is defined early using only $\ker(\mathbf{M})$ -compatibility (Layer Q), A_{phy} is defined late using fixed-point closure (Layer Q'), and Theorem 0.7.3 certifies the split is harmless under the stable-mode assumption.

Lemma 0.6.2 — Thin Candidate is a Sub-Algebra

LEMMA 0.6.2 — Thin Candidate is a Sub-Algebra [Established]

LEMMA 0.6.2 (Thin Candidate is a Sub-Algebra).

$A_{\text{phy}}^{\text{thin}}$ is a unital $(*)$ -sub-algebra of A .

Proof. Let $A, B \in A_{\text{phy}}^{\text{thin}}$ and $\lambda \in C$. Then:

- Closure under addition: $\pi_{\text{kin}}(A+B) \ker(M^{\wedge}) = \pi_{\text{kin}}(A) \ker(M^{\wedge}) + \pi_{\text{kin}}(B) \ker(M^{\wedge}) \subseteq \ker(M^{\wedge})$, so $A+B \in A_{\text{phy}}^{\text{thin}}$.
- Closure under multiplication: $\pi_{\text{kin}}(AB) \ker(M^{\wedge}) = \pi_{\text{kin}}(A) [\pi_{\text{kin}}(B) \ker(M^{\wedge})] \subseteq \pi_{\text{kin}}(A) \ker(M^{\wedge}) \subseteq \ker(M^{\wedge})$, so $AB \in A_{\text{phy}}^{\text{thin}}$.
- Closure under involution: follows from $P^{\wedge}_0$ self-adjointness and $[A, P^{\wedge}_0] = 0$ on $\ker(M^{\wedge})$ implying $[A^{\dagger}, P^{\wedge}_0] = 0$ on $\ker(M^{\wedge})$ (taking adjoints).
- Unit: $\pi_{\text{kin}}(1) \ker(M^{\wedge}) = \ker(M^{\wedge})$, so $1 \in A_{\text{phy}}^{\text{thin}}$. $\square$

§0.7 The Full Physical Sub-Algebra and Theorem 0.7.3 (Layer Q' — Quartic)

LAYER Q' · QUARTIC

Source: Section_0_2.pdf §0.6 (Definition 0.6.1), **REWRITE per Construction Spec Ch. 9** (Theorem 0.6.3 restored from Front.pdf §0.6.3, where it was silently dropped in Gen 3). **Epistemic tag:** [Theorem Target] — the thin/full equality depends on the stable-mode assumption (shared with Theorem 4.2.2 and Theorem Target T-2).

The full physical sub-algebra $\mathbf{A}_{\text{phy}}$ is the fixed-point algebra of the asymptotic Reconciliation Propagator $\mathbf{R}_{\infty}$: it is the algebra of operators that commute with the asymptotic projector in the strongest sense. This is a Quartic-layer object: it requires the propagator to have been composed (Layer Q) and the Reconciliation Principle to have been stated (§0.8). Theorem 0.7.3 — silently dropped in Gen 3 and restored here per Spec Ch. 9 — proves that the thin candidate of §0.6 and the full algebra of §0.7 coincide under the stable-mode assumption, certifying that the thin/full split is harmless.

Definition 0.7.1 — Full Physical Sub-Algebra

DEFINITION 0.7.1 — Full Physical Sub-Algebra [Established]

DEFINITION 0.7.1 (Full Physical Sub-Algebra).

The full physical sub-algebra is the fixed-point algebra of the asymptotic Reconciliation Propagator $\mathbf{R}_{\infty} = \lim_{t \rightarrow \infty} \mathbf{R}_t$:

$$\mathbf{A}_{\text{phy}}^{\text{full}} = \{ A \in \mathbf{A} : [\mathbf{A}, \mathbf{R}_{\infty}] = 0 \} \quad (0.7.1)$$

Equivalently: $A \in \mathbf{A}_{\text{phy}}^{\text{full}}$ iff A commutes with the asymptotic projector $\mathbf{P}_{\infty}$ AND preserves the full $\mathbf{F}$ -spectral decomposition of $\ker(\mathbf{M}^{\wedge})$ (not just the kernel itself).

This is a Layer Q' (Quartic) object — it requires $\mathbf{R}_t$ to have been composed (Layer Q) and the Reconciliation Principle to have been stated (§0.8).

Theorem 0.7.3 — Fixed-Point Characterization (Restored)

Why this theorem was restored

Theorem 0.6.3 (renumbered 0.7.3 in the merged canonical form) appeared in the Gen 1 drafts (Front.pdf §0.6.3, the unnumbered .docx) and was silently dropped in Gen 3. Without it, the thin/full split of §0.6/§0.7 is unjustified — readers cannot verify that the early candidate deserves the load-bearing role Sections 1–2 give it. Restoring the theorem is the certification step that the restructuring actually worked. (Construction Spec Ch. 9.)

THEOREM 0.7.3 — Fixed-Point Characterization [Theorem Target T-2] (restored per Ch. 9)

THEOREM 0.7.3 (Fixed-Point Characterization).

$$\text{Let } A_{\text{phy}}^{\text{thin}} = \{ A \in A : \pi_{\text{kin}}(A) \ker(M^{\wedge}) \subseteq \ker(M^{\wedge}) \} \quad (\$0.6)$$

$$\text{and } A_{\text{phy}}^{\text{full}} = \{ A \in A : [A, R_{\infty}] = 0 \} \quad (\$0.7).$$

Under the stable-mode assumption on the reconciliation flow,

$$A_{\text{phy}}^{\text{thin}} = A_{\text{phy}}^{\text{full}} \quad (0.7.3)$$

Equivalently: $[A, R_{\infty}] = 0$ iff $A \in A_{\text{phy}}^{\text{thin}}$ and A restricted to $\ker(M^{\wedge})$ preserves the spectral decomposition induced by $F^{\wedge}$.

Proof sketch.

- ($\subseteq$) If $[A, R_{\infty}] = 0$ then A commutes with the asymptotic projector $P^{\wedge}_0$ onto $\ker(M^{\wedge})$. Hence $\pi_{\text{kin}}(A)$ preserves $\ker(M^{\wedge})$, giving $A \in A_{\text{phy}}^{\text{thin}}$.
- ($\supseteq$) Conversely, if $A \in A_{\text{phy}}^{\text{thin}}$, the stable-mode assumption implies A 's action on the $\ker(M^{\wedge})$ decomposition is invariant under R_t 's flow. By the Master-Zero equality $\ker(M^{\wedge}) = \ker(F^{\wedge})$ (Corollary 0.5.2), this invariance extends to the limit R_{∞} , giving $[A, R_{\infty}] = 0$. $\square$

Remark. The stable-mode assumption is the same one used in Theorem 4.2.2 (mass-burden spectral equivalence, merged Section 4). The two theorems share a common hypothesis, which is the framework's central technical conjecture (Theorem Target T-2).

STATUS: [Theorem Target]. Closure depends on T-2 (spectral gap analysis of R_t on $\ker(M^{\wedge})$).

§0.7b The Emergent Dirac Bracket (Layer Q' — Quartic)

LAYER Q' · QUARTIC

Source: Section_0_2.pdf §0.6 Theorem 0.6.2 (Emergent Dirac Bracket), paired with Front.pdf §0.4.3 Definition 0.8 (Algebraic Dirac Bracket). *Per Construction Spec Ch. 9:* was dropped in Gen 3; restored here. *Epistemic tag:* [Conditional Theorem / Theorem Target].

The Dirac bracket appears in the framework in two complementary forms. Front.pdf §0.4.3 defines it *algebraically*, as the modified commutator that resolves the Tier 1 second-class structure by inverting the compatibility matrix. Section_0_2 Theorem 0.6.2 derives it *dynamically*, as the fixed-point effective commutator of the MOE descent — operators that survive global burden minimization satisfy the Dirac bracket as their asymptotic commutator. The two forms should be paired: the algebraic definition is established; the dynamical derivation is conditional. The Gen 3 draft dropped the dynamical form; this merged section restores it.

Definition 0.7b.1 — Algebraic Dirac Bracket (Front.pdf)

DEFINITION 0.7b.1 — Algebraic Dirac Bracket [Established] (Front.pdf §0.4.3)

DEFINITION 0.7b.1 (Algebraic Dirac Bracket).

Let $\{K^a\}_{a \in I_1}$ be the Tier 1 (second-class) primitive constraints, with invertible compatibility matrix

$\Delta^{(1)}_{ab} = [K^a, K^b]$. The algebraic Dirac bracket of

$A, B \in A$ is

$$[A, B]_D = [A, B] - \sum_{a,b} [A, K^a] \cdot (\Delta^{(1)})^{ab} \cdot [K^b, B] \quad (0.7b.1)$$

where $(\Delta^{(1)})^{ab}$ is the inverse of $\Delta^{(1)}_{ab}$.

LEMMA 0.7b.2. $[A, K^c]_D = 0$ for all $A \in A$ and all $c \in I_1$.

(The Dirac bracket annihilates Tier 1 constraints.)

The reduced algebra $A_{\text{red}} := A / I_K$ is the quotient by the Tier 1 ideal, equipped with $[\cdot, \cdot]_D$ as its internal commutator.

Theorem 0.7b.3 — Emergent Dirac Bracket (Section_0_2)

THEOREM 0.7b.3 — Emergent Dirac Bracket [Theorem Target T-2] (restored per Ch. 9)

THEOREM 0.7b.3 (Emergent Dirac Bracket as MOE Fixed-Point).

The Dirac bracket $[\cdot, \cdot]_D$ is the fixed-point effective commutator of the MOE descent. Operators that survive global burden minimization satisfy

$$[A, B]_{\text{eff}} = [A, B] - \sum_{\{a,b\}} [A, K^a] \cdot (\Delta^{\{ab\}}) \cdot [K^b, B] \quad (0.7b.3)$$

where K^a are the dressed constraint representatives generated by within-extension SOE spectral flow, and $(\Delta^{\{ab\}})$ is the inverse of the dressed compatibility matrix.

Mechanism:

- Within-extension SOE spectral flow generates the dressed constraint representatives K^a
- Cross-extension MOE descent ensures they are globally valid
- The fixed-point commutator coincides with the algebraic Dirac bracket (Definition 0.7b.1)

STATUS: [Conditional Theorem / Theorem Target]. The full proof requires the stable-mode assumption (Theorem Target T-2).

Interpretation: the Dirac bracket is not an imported Dirac-Bergmann algorithm. It is the asymptotic commutator of the reconciliation flow itself. The algebraic definition (Front.pdf) and the dynamical derivation (Section_0_2) are two sides of the same theorem.

§0.8 The Reconciliation Principle (Layer Q' — Quartic)

LAYER Q' · QUARTIC

*Source: Ddd.pdf §1.2 (variational form), Front.pdf §1.5.2 (Gen 1 placement). **RELOCATED per Construction Spec Ch. 6:** removed from §0.4 (Gen 3 placement, too early); restated here in canonical form immediately after the Cubic ingredients of Sections 1–2 become available. **Epistemic tag:** [Structural Principle].*

The Reconciliation Principle (RP) is the dynamical engine of the framework — the only place where "change" originates. Everything else (time, mass, algebras, propagation) is an effect of its minimization. In the canonical manuscript, RP must be stated as a Layer Q' (Quartic) principle, immediately after Section 2 has supplied its Cubic ingredients (causal order $<$, correlation kernel K_ω). The Gen 3 placement (§0.4) is too early; the Gen 1 Front.pdf placement (§1.5.2) is correct in spirit but uses pre-SOE/MOE language. The canonical placement fuses Gen 1's ordering with Gen 3's machinery.

Forward-reference contract (resolved when Sections 1–2 merge)

The RP variational target $I(S)$ requires two Cubic-layer ingredients: (i) the set S of mutually correlated events, defined via the correlation kernel K_ω (merged Section 2.1), and (ii) the causal order $<$ used to filter unordered pairs (merged Section 1.1). In this merged Section 0 these appear as forward references. The principle is stated in its final canonical form; its full minimization target becomes concrete when Sections 1 and 2 are merged. No circularity is introduced: §0.1–0.7b do not depend on §0.8 — RP is the *output* of the foundation, not its input. (Construction Spec Ch. 6.)

0.8.1 Canonical Statement

§0.8 — RELOCATED Reconciliation Principle [Structural Principle]

§0.8 THE RECONCILIATION PRINCIPLE (Layer Q' — Quartic)

Prerequisites (all available at this point in the manuscript):

- (i) Set S of mutually correlated events (Sec 2.1, K_ω)
- (ii) Causal order $<$ to filter unordered pairs (Sec 1.1)
- (iii) Reconciliation Propagator R_t (Sec 0.4, Layer Q)
- (iv) Master-Zero $\ker(M^\wedge) = \ker(F^\wedge)$ as attractor (Sec 0.5)

PRINCIPLE (Reconciliation, variational form).

For any set S of mutually correlated events, the system evolves under R_t toward a state that minimizes the total relational inconsistency

$$I(S) = \sum_{\{a,b \in S, a \nrightarrow b, b \nrightarrow a\}} [s(a,b) - s^*(a,b)]^2 \quad (0.8.1)$$

where $s(a,b)$ is the current correlation strength and $s^*(a,b)$ is the unique correlation strength consistent with all causal constraints impinging on a and b .

THEOREM 0.8.2 (Well-Posedness).

Under the stable-mode assumption, the minimizer of $I(S)$ exists and is unique. It coincides with the asymptotic fixed point of R_t , i.e., with $\ker(M^\wedge) = \ker(F^\wedge)$.

COROLLARY 0.8.3 (Fracture Event).

Where the minimizer is non-trivial (multiple stable sub-sets), the kinematic Hilbert space H_{kin} fractures into sector sub-spaces $H_{\{\omega,k\}}$ invariant under R_∞ . The sector sub-algebras

$$A_k = \{ A \in A_{\text{phy}} : [\pi_{\text{kin}}(A), P_k] = 0 \}$$

are DERIVED at this point, not assumed. They are outputs of RP, not inputs.

0.8.2 Migration Notes (per Construction Spec Ch. 6)

Three migration actions are encoded in the relocated statement. First, any statement of RP in §0.4 of the Gen 3 draft is deleted — the SOE flow equation there has been re-expressed without borrowing Section 1 vocabulary (per the $\delta\hat{C}$ replacement of §0.4). Second, the Ddd.pdf §1.2 statement of the variational minimization is ported verbatim into §0.8 above; it is the cleanest formulation in the archive. Third, the fracture theorem (Corollary 0.8.3) is phrased as a derivation from RP, not as a parallel postulate — this is the joint that resolves the structural Issue 1 (sub-algebras defined too early).

§0.9 Architectural Summary

The merged Section 0 is now closed. Every definition sits at the layer where its ingredients first exist; every theorem either has a proof sketch or is explicitly flagged as a Theorem Target; the two structural regressions of Gen 3 (sub-algebra ordering, RP placement) are resolved by the thin/full split and the §0.8 relocation respectively. The emergence ladder is acyclic: no layer depends on a layer below it, and no definition is used

before its ingredients exist.

§	Structure	Mechanism	Layer	Status
0.1	Kinematic algebra $\mathbf{A}$, primitive constraints, $\mathbf{M}$	—	L	Definition
0.2	Kinematic state ω_{kin} , constraint ideal I_C , compatibility bracket $\Delta_{\alpha\beta}$	—	L	Definition
0.3	GNS Hilbert space $\mathbf{H}_{\text{kin}}$, fracture operator $\mathbf{F}$, burden $\mathbf{B}_\Delta$, Π_{net}	—	Q	Established
0.4.1	SOE spectral-flux flow (local, isometric)	Flux redistribution + unitary eigenbasis rotation	Q	Conditional
0.4.2	MOE gradient descent (global, contractive)	Bures descent $\rightarrow$ Lindblad leading order	Q	Conditional + T-2
0.4.3	Dephasing (cross-eigenspace suppression)	Residual MOE-scale effect	Q	Conditional
0.5	Convergence Theorem: $\mathbf{H}_{\text{kin}} \rightarrow \ker(\mathbf{M}) = \ker(\mathbf{F})$	SOE + MOE composition	Q	Conditional (T-2 for ∞ -dim)
0.6	Thin candidate $\mathbf{A}_{\text{phy}}^{\text{thin}}$ ($\ker(\mathbf{M})$ -compatibility)	Spectral compatibility alone	Q	Established
0.7	Full $\mathbf{A}_{\text{phy}}$ (fixed-point of $\mathbf{R}_\infty$)	Fixed-point closure	Q'	Definition
0.7.3	Theorem 0.7.3: $\mathbf{A}_{\text{phy}}^{\text{thin}} = \mathbf{A}_{\text{phy}}^{\text{full}}$	Stable-mode assumption	Q'	Theorem Target T-2
0.7b	Emergent Dirac bracket as MOE fixed-point	SOE dressing + MOE descent	Q'	Theorem Target T-2
0.8	Reconciliation Principle (variational)	Minimization of $I(S)$	Q'	Structural

Table 0.9.1 — Architectural summary of merged Section 0. 12 structural units across 4 layers; no forward references.

The Reconciliation Propagator is now a THREE-MECHANISM composite: SOE spectral-flux flow (local, isometric) + MOE gradient descent (global, contractive) + dephasing (residual). The Three-Layer Protocol assigns QM structure to SOE (Layer B) and GR structure to MOE (Layer C), with the Convergence Theorem providing dynamical justification for both. Master-Zero is derived, not postulated. The physical sub-algebra is split into a thin candidate (defined early, sufficient for Section 1.1.1) and a full algebra (defined late, certified equal by Theorem 0.7.3). The Reconciliation Principle is the output of the foundation, not its input.

What this Section unlocks for the next merge

With Section 0 closed, Section 1 (Causal Foundation) can be merged against a stable foundation. Section 1.1.1 (event) will be restated to require only the thin candidate $\mathbf{A}_{\text{phy}}^{\text{thin}}$ from §0.6 — no forward reference to §0.7 or §0.8. The causal order $<$ introduced in Section 1.1 feeds back into §0.8 as one of the two Cubic ingredients of the RP variational target. Section 2 (Emergent Space) supplies the other ingredient (correlation kernel K_{ω}), completing the RP target and grounding the fracture event of Corollary 0.8.3. The merger order Section $0 \rightarrow 1 \rightarrow 2$ is therefore not arbitrary; it is the order in which the Cubic layer is populated, allowing the Quartic-layer principle to become fully operational.